Documentation

System Name: Employee Management System (EMS)

Team members:

- Mazen Amged Afifi Abdelhamid(Team leader)
- Nada yehia abdelaziz abo elala
- Aya Ahmed Abd ElShafy Ibrahim Helal
- Radwa Elsayed Ahmad

1. Project planning and management:

- **Project Description:**The project is a digital system designed to manage employee attendance, lateness, absences, leave requests, and salary deductions within companies. It uses GPS-based geofencing to ensure that employees can register their attendance only when they are inside the company's designated work area. The system also provides automated attendance tracking, early-arrival compensation rules, and leave approval workflows to reduce paperwork and improve administrative efficiency.

- **Project objectives:**
  1-Replace traditional paper-based attendance records and fingerprint systems.
  2-Track employee attendance, lateness, and absences automatically.
  3-Verify employee location during attendance registration.
  4-Automate leave requests and manager approvals.
  5-Calculate deductions according to company policies.
  6-Provide manager with reports about employees' attendance and leave records.
  7-Reduce manual errors and save time for HR department.

- **Proposed Timeline:**
    1) **W1 — Week 1: Project Setup & UI Design**
    2) **W2 — Week 2: UI Implementation**
    3) **W3 — Week 3: Navigation & Mock Data**
    4) **W4 — Week 4: Local Storage & Basic Logic**
    5) **W5 — Week 5: API Fundamentals & Integration**
    6) **W6 — Week 6: Backend & Database Integration**
    7) **W7 — Week 7: Location & Business Rules**
    8) **W8 — Week 8: Integration, Testing & Delivery**

- **Team Responsibilities:**
    1) **Aya — Project Coordination & Integration**
        - ❖ Organize the GitHub repository, branches, and pull requests.
        - ❖ Track project progress and ensure tasks follow the project timeline.
        - ❖ Set up the project structure and navigation in collaboration with the team.
        - ❖ Coordinate API integration and data models across different features.
        - ❖ Assist with feature integration and resolving integration issues.

    2) **Mazen — Employee Attendance**
        - ❖ Develop the employee home screen and attendance history interface.
        - ❖ Implement check-in and check-out functionality.
        - ❖ Apply working-hour rules and calculate lateness.
        - ❖ Implement location verification to ensure employees are within the company's permitted work area.
        - ❖ Integrate attendance features with the API when it becomes available.

    3) **Nada — Admin & Attendance Management**
        - ❖ Develop the Admin dashboard and employee list.
        - ❖ Display employee attendance, check-in/check-out times, lateness, and absences.
        - ❖ Implement attendance record management according to assigned permissions.
        - ❖ Integrate administrative features with the API.
        - ❖ Verify that employee attendance records are displayed correctly.

4) **Radwa — Leave, Bonuses & Deductions Management**
   - ❖ Develop leave request forms and leave status tracking.
   - ❖ Implement leave approval and rejection by the Admin.
   - ❖ Implement bonus and deduction management.
   - ❖ Develop company holiday management.
   - ❖ Integrate these features with the data source and test record update.

Shared Responsibilities

All team members will collaborate on the following tasks:

- *UI/UX Design:* Contribute to designing a consistent and user-friendly interface.
- *Learning & Technical Development:* Learn and apply the required API and local storage concepts.
- *Backend & Database Integration:* Collaborate on data integration, with specific responsibilities assigned according to the agreed architecture.
- *Code Review & GitHub Collaboration:* Review code, manage changes, and follow agreed coding standards.
- *Testing & Bug Fixing:* Test individual features, identify bugs, and verify the complete application.
- *Documentation & Final Presentation:* Contribute to project documentation and prepare the final presentation.
- *Final Integration & Delivery:* Work together to integrate all features and prepare the final application, coordinated by Member 1.

Implementation Approach

Each member will develop their assigned features incrementally:

1. Build the user interface using mock data.
2. Implement the basic business logic.
3. Apply the required API or local storage concepts.
4. Connect the feature to the actual data source when available.
5. Test the feature and integrate it with the rest of the application.

This approach allows the team to work in parallel, learn unfamiliar technologies gradually, and maintain consistent integration throughout development

- **Risk Management :**
  - ❖ **GPS Inaccuracy:**Set an acceptable accuracy threshold and verify the employee's location before registering attendance.
  - ❖ **Location Spoofing:**Use server-side verification and mechanisms to detect fake GPS locations.
  - ❖ **Employee Account Theft:**Implement secure authentication and appropriate access permissions.
  - ❖ **Incorrect Deduction Calculations:**Document company policies clearly and test different calculation scenarios.
  - ❖ **Internet Connection Failure:**Display a clear error message and do not confirm attendance until it has been verified by the server.
  - ❖ **Unauthorized Data Access:**Encrypt sensitive data and enforce role-based access control.
- **Success Metrics:**
  - ➢ Recording attendance in the shortest possible time.
  - ➢ Reducing manual errors in attendance records.
  - ➢ Calculating lateness and deductions in accordance with company policy.
  - ➢ Successful review of leave requests by managers via the application.
  - ➢ Preventing attendance recording from outside authorized locations, ensuring GPS accuracy.
  - ➢ Providing clear and accurate reports for Human Resources.

2. Literature Review

Time and Attendance systems are the cornerstone of Human Resources (HR) digital transformation, aiming to minimize human error, automate payroll calculations, and enhance overall productivity. This review examines leading global systems in this domain;renowned for GPS location tracking ;recognized for task scheduling and team communication ;known for its automated leave request workflows ;

Despite the significant advancements in similar systems, this review reveals a distinct Research Gap in the market. This gap is characterized by three main factors:

(1) High subscription costs and rigid penalty structures.

(2) A primary focus on "monitoring" rather than "motivation" (i.e., the absence of a feature to compensate for lateness with early arrival).

(3) A lack of real-time financial and time-related transparency available to the employee.

Consequently, the current project introduces an innovative solution that breaks these traditional, rigid rules. It designs a system that utilizes GPS Geofencing for location verification, integrated with a unique Smart Penalty & Reward Engine. This engine introduces the concept of "Time Banking," allowing employees to offset their lateness with early arrivals. Furthermore, the system provides a comprehensive dashboard ensuring Absolute Transparency for both the employee and the administration, fostering a workplace built on Organizational Justice and significantly reducing administrative disputes.

3. Requirements Gathering

3.1 Stakeholders and User Needs

The proposed system is designed to simplify employee attendance management by replacing traditional paper-based records with a centralized digital solution. The system has two primary user roles:

1. Employee

Employees need a simple and reliable way to manage their daily attendance and view their attendance-related records. Their needs include:

- Registering check-in and check-out times based on their location relative to the company's designated work area.
- Viewing attendance records, including on-time attendance, lateness, and absences.
- Tracking accumulated bonuses and recorded deductions.
- Viewing their attendance and leaving times.
- Submitting leave requests and tracking their approval status.

2. Admin

The Admin is responsible for supervising employee attendance and managing attendance-related records according to company policies. Their needs include:

- Monitoring employee attendance, lateness, and absences.
- Managing attendance records and reviewing employee check-in and check-out times.
- Adding bonuses and modifying or removing deductions according to their permissions.
- Managing holidays and leave requests, including accepting or rejecting requests.

- Accessing organized employee attendance records to support administrative decisions.

3.2 Functional Requirements

Functional requirements describe the services and operations that the system must provide.

A. Employee Features

FR-01: Attendance Registration

The system shall allow employees to register their check-in and check-out times through the application.

FR-02: Location Verification

The system shall verify the employee's current location and allow attendance registration only when the employee is within the permitted radius of the company's designated work area.

FR-03: Attendance Status

The system shall classify attendance records according to the company's configured working schedule, distinguishing between on-time attendance and late arrival.

FR-04: Attendance History

The system shall allow employees to view their recorded attendance, check-in and check-out times, and attendance status.

FR-05: Bonuses and Deductions

The system shall allow employees to view their accumulated bonuses and recorded deductions.

FR-06: Leave Requests

The system shall allow employees to submit leave requests, provide a reason, and track the status of each request.

B. Admin Features

FR-07: Attendance and Absence Management

The system shall allow the Admin to monitor employee attendance, lateness, and absences and manage attendance records according to the system's permissions.

FR-08: Bonus Management

The system shall allow the Admin to add bonuses to employee records and maintain the corresponding records.

FR-09: Deduction Management

The system shall allow the Admin to review, modify, or remove recorded deductions according to company policies and assigned permissions.

FR-10: Leave and Holiday Management

The system shall allow the Admin to review leave requests, accept or reject them, and manage company holidays.

FR-11: Attendance Records Overview

The system shall provide the Admin with access to employee attendance records to support monitoring and administrative follow-up.

C. System Rules

FR-12: Working Schedule Configuration

The system shall apply the company's configured working hours when determining attendance status.

FR-13: Location Radius Configuration

The system shall use the permitted geographical radius defined for the company's work area when validating attendance registration.

FR-14: Role-Based Access

The system shall restrict available features and operations according to the user's role, ensuring that employees can access their own records while administrative operations remain restricted to authorized users.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational standards expected from the system.

NFR-01: Security

The system shall protect employee information and attendance records through secure authentication, authorization, and role-based access control.

NFR-02: Performance

The system should process attendance registration and retrieve attendance records within a reasonable response time under normal operating conditions.

NFR-03: Usability

The application shall provide a clear and user-friendly interface that enables employees and administrators to perform their tasks with minimal effort.

NFR-04: Reliability

The system shall maintain consistent attendance records and prevent unauthorized or duplicate attendance entries for the same attendance event.

NFR-05: Data Integrity

The system shall preserve the accuracy and consistency of attendance, bonus, deduction, and leave records. Changes to existing records should be restricted to authorized users.

NFR-06: Maintainability

The system should be organized into modular components to facilitate future updates, bug fixes, and the addition of new features.

NFR-07: Compatibility

The employee application shall support the target mobile platforms selected for the project and operate on compatible devices with the required location services.

NFR-08: Privacy

The system shall use employee location information for attendance verification and handle personal data only for authorized system purposes.

NFR-09: Availability and Connectivity

The system should handle temporary connectivity problems appropriately and provide clear feedback when an operation cannot be completed. Successful attendance registration must be confirmed by the system.

3.4 System Scope and Constraints

Project Scope

The initial version of the system will focus on:

- Employee check-in and check-out using location-based verification.
- Attendance, lateness, and absence tracking.
- Employee access to attendance history, bonuses, and deductions.
- Administrative management of attendance records, bonuses, deductions, holidays, and leave requests.
- Applying configurable company working hours and location-radius rules.
- Providing role-based access to employee and administrative features.

Project Constraints

- Location-based attendance depends on device location permissions, GPS accuracy, and the availability of location services.
- The system requires an internet connection for server-side attendance validation and synchronization of records.
- Working hours, permitted location radius, and attendance policies must be configured according to the company's rules.
- Bonus and deduction management will follow the policies defined for the project; the system will not necessarily perform complete payroll processing.
- The initial implementation will be limited to the features agreed upon by the project team and the available development time.

3.5 Assumptions

- Each employee will have an authorized account to access the system.
- The Admin will be authorized to manage employee attendance-related records.
- The company will provide its designated work location, permitted radius, working hours, and relevant attendance policies.
- Employees will grant the required location permissions when registering attendance.
- Leave approval, holiday management, bonuses, and deductions will follow the rules configured for the company.